Lecture #3: Independence & Conditional Probability

MATH 321 – Mathematical Statistics I

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Formulas for $P(A \cup B)$ and $P(A \cap B)$

Recall that:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B),$$

and if A and B are mutually exclusive, then $P(A \cap B) = 0$ and:

$$P(A \cup B) = P(A) + P(B).$$

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and if A and B are mutually exclusive, then $P(A \cap B) = 0$ and:

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But a formula for $P(A \cap B)$ when the events are not mutually exclusive is not as simple, we will need to develop some new concepts.

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Independence

Two events are **independent** if the occurrence of one event does not affect the probability of the other event occurring.

In our formal notation, that is:

$$P(A \cap B) = P(A) \times P(B)$$

Independence occurs when knowing information about one event or random variable does not give us any information about the other event or random variable.

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Independence Examples

1. What is the probability of rolling two 1s on a six-sided die?

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2. What is the probability of seeing ten heads when flipping ten fair coins?

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Independence Examples

1. What is the probability of rolling two 1s on a six-sided die?
2. What is the probability of seeing ten heads when flipping ten fair coins?
3. Can two events be *both* mutually exclusive *and* independent?

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Non-Independence

Most events are not independent, however. We might try to assume independence for many statistical operations, but that's not always a sound assumption to make.

How then do we calculate $P(A \cap B)$ if we can't assume independence?

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Non-Independence

Most events are not independent, however. We might try to assume independence for many statistical operations, but that's not always a sound assumption to make.

How then do we calculate $P(A \cap B)$ if we can't assume independence?

Example: Suppose two cards are drawn from a deck without replacement. What is the probability that both are spades?

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Conditional Probability

The **conditional probability of A given B**, denoted by $P(A \mid B)$, is defined by

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad \text{provided } P(B) > 0.$$

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This means that, in general:

$$P(A \cap B) = P(A \mid B) \times P(B) = P(B \mid A) \times P(A)$$

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Example (Again): Suppose two cards are drawn from a deck without replacement. What is the probability that both are spades?

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Conditional Probability Example

Example: E-mail sent to a person can either be "spam" or not "spam". It can also contain the word "offer" or not contain the word. Suppose 160 emails are sampled and the following table is computed:

	Spam	**Not Spam**
"Offer"	40	5
No "Offer"	15	100

1. What is the probability an e-mail contains the word "offer" given that it is spam?

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Conditional Probability Example

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	Spam	**Not Spam**
"Offer"	40	5
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1. What is the probability an e-mail contains the word "offer" given that it is spam?
2. What is the probability an e-mail doesn't contain the word "offer" given that it is spam?

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Conditional Probability Example

Example: Gregor Mendel was a monk who, in 1865, suggested a theory of inheritance based on the science of genetics. He identified heterozygous individuals for flower color that had two alleles (one r = recessive white color allele and one R = dominant red color allele). When these individuals were mated, $\frac{3}{4}$ of the offspring were observed to have red flowers, and $\frac{1}{4}$ had white flowers.

The following table summarizes this mating; each parent gives one of its alleles to form the gene of the offspring.

		Parent 2	
Parent 1		$\boldsymbol{r}$	$\boldsymbol{R}$
	$\boldsymbol{r}$	rr	rR
	$\boldsymbol{R}$	Rr	RR

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Conditional Probability Example (Continued)

We assume that each parent is equally likely to give either of the two alleles and that, if either one or two of the alleles in a pair is dominant (R), the offspring will have red flowers. What is the probability that an offspring has:

1. at least one dominant allele?
2. at least one recessive allele?
3. one recessive allele, given that the offspring has red flowers?

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The Law of Total Probability

The Law of Total Probability:

$$P(A) = P(A \mid B) \times P(B) + P(A \mid \overline{B}) \times P(\overline{B})$$

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$$P(A) = P(A \mid B) \times P(B) + P(A \mid \overline{B}) \times P(\overline{B})$$

This can be generalized. If $B_1, B_2, B_3, \ldots$ form a **partition** of the sample space, then:

$$P(A) = \sum_{i \in I} \left[P(A \mid B_i) \times P(B_i) \right]$$

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Partition

For some positive integer k, let the sets $B_1, B_2, \ldots, B_k$ be such that:

1. $\Omega = B_1 \cup B_2 \cup \cdots \cup B_k$.
2. $B_i \cap B_j = \varnothing$, for $i \neq j$.

Then the collection of sets $\{B_1, B_2, \ldots, B_k\}$ is said to be a **partition** of Ω.

The Law of Total Probability also holds if the partition consists of a countably infinite collection of sets.

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The Law of Total Probability Example

A population of voters contains 40% Republicans and 60% Democrats. It is reported that 30% of the Republicans and 70% of the Democrats favor an election issue. What is the probability that a person chosen at random from this population favors the issue?

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Bayes' Theorem

Bayes' Theorem:

If we want to "flip" the events in a conditional probability, we can thank Thomas Bayes (1701–1761).

$$P(A \mid B) = \frac{P(B \mid A) \times P(A)}{P(B)}$$

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Bayes' Theorem

We can combine this with the Law of Total Probability, to get the formulation:

$$P(A \mid B) = \frac{P(B \mid A) \times P(A)}{P(B \mid A) \times P(A) + P(B \mid \overline{A}) \times P(\overline{A})}$$

or:

$$P(A_j \mid B) = \frac{P(B \mid A_j) \times P(A_j)}{\sum_{i \in I} \left[P(B \mid A_i) \times P(A_i) \right]}$$

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Bayes' Theorem Example

Example (Voters, Revisited): A population of voters contains 40% Republicans and 60% Democrats. It is reported that 30% of the Republicans and 70% of the Democrats favor an election issue. A person chosen at random from this population is found to favor the issue in question. What is the probability that this person is a Democrat?

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Bayes' Theorem Example

Example: An airport explosives detector lights up with probability 0.75 when a piece of luggage contains an explosive, but also lights up with probability 0.01 when it doesn't. Suppose that a piece of luggage has only a 0.0001 chance of containing an explosive. If the detector lights up, what is the probability the luggage contains an explosive?

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Bayes' Theorem Example

Example: Suppose there is a genetic disorder that occurs in 1 in every 100000 births. A team of researchers develop a test to see if you have it.

If you have the disorder, there is a 99.9% chance that the test will return a positive result.

If you do not have the disorder, there is a 99.95% chance that the test will return a negative result.

You have the test run on yourself, and you get a positive result. What is the probability that you have the disorder?

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